

Uniform IDS approximation for algebraic local coefficients

A partial answer to the arbitrary-coefficient question in arXiv:1304.0945

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Status. Candidate partial result; current verdict: *likely valid*. Human review is recommended. The theorem covers every fixed finite set of real algebraic coefficients, but it does not settle genuinely transcendental coefficients.

1 Source question

The source is Felix Pogorzelski, *Convergence theorems for graph sequences*, arXiv:1304.0945, Section 6.2, page 12 [3]. For a weakly convergent bounded-degree graph sequence and a fixed finite-range pattern-invariant selfadjoint operator sequence, the source recalls pointwise convergence of spectral atom masses for integer-valued local kernels (Theorem 6.10). It then asks whether the resulting pointwise, equivalently uniform, convergence remains true for arbitrary coefficients.

Transcription. After deriving uniform convergence for bounded integer-coefficient operators, the source states: “To the knowledge of the author, this is still an open question for arbitrary coefficients.”

2 Set-up

Let (G_n) be finite graphs of maximum degree at most d , with $|V(G_n)| \rightarrow \infty$, which converge weakly (in the Benjamini–Schramm sense). Let H_n be selfadjoint matrices arising from one fixed pattern-invariant kernel rule of range R : every row is determined by the rooted R -ball, and the rule does not change with n . There are only finitely many rooted R -ball types and only finitely many vertices in each such ball, so the set S of all matrix-entry values used by this rule is finite.

Write

$$N_n = |V(G_n)|, \quad \nu_n = \frac{1}{N_n} \sum_{j=1}^{N_n} \delta_{\lambda_{n,j}}.$$

Weak graph convergence and locality imply that all moments of ν_n converge. Uniform boundedness of the matrices makes the supports compact, so ν_n converges weakly to the source paper’s density-of-states measure ν ; this is source Theorem 6.6 [3].

3 Idea of the proof

For integer matrices, the Abért–Thom–Virág argument uses the fact that a nonzero product of eigenvalue differences is a nonzero integer. Hence it cannot be too small. With algebraic entries,

It can already be noticed in the group situation that one cannot expect the Banach space convergence of Theorem 4.2 to hold for non-amenable structures. Indeed, it has been shown in [12] (Proposition 4.1) that sofic approximations of groups are hyperfinite if and only if the group under consideration is amenable. Thus, we infer from Proposition 2.8 that our notion of almost-additive functions will not allow for normalized convergence along approximating sequences in *non-amenable* groups. However, as far as the question of pointwise (or uniform) convergence of spectral distribution functions is concerned, one can bypass this problem by extending the algebraic eigenvalue conjecture (see e.g. [5]) to the general geometric case of graph sequences. This issue has been addressed by ABÉRT, THOM and VIRÁG in [2]. The method is to use the number theoretic techniques developed in [31] for the proof of the conjecture in the case of sofic groups. Precisely, if ν_n and ν are the positive measures associated with N_{H_n} and $N_{[\mathcal{H}]}$ respectively, then the following assertion holds true.

Theorem 6.10 (cf. [2]). *Let $(G_n) \subseteq \mathcal{G}_d$ be a weakly convergent graph sequence. Assume that $[\mathcal{H}] := (H_n)$ is an associated weakly convergent operator sequence with integer coefficients (admissible kernel functions), i.e. for all $\alpha \in \mathcal{A}_d$, we have that h_α takes its values in $\mathbb{Z}$. Then,*

$$\lim_{n \rightarrow \infty} \nu_n(\{\lambda\}) = \nu(\{\lambda\}) \quad (6.1)$$

for every $\lambda \in \mathbb{R}$.

Now well-known arguments can be used to derive from the limit relation (6.1) the uniform convergence of (N_{H_n}) , see e.g. [21], Lemma 6.3. Thus, one obtains the uniform existence of the integrated density of states for bounded, integer coefficient operators on all approximable, discrete structures with vertex degree bound. To the knowledge of the author, this is still an open question for arbitrary coefficients, see the remark following Theorem 4.4 in [31].

These elaborations show that strong spectral approximation results hold true even if one does not have a Banach space valued convergence theorem at hand. However, one may raise the question whether one can find a class of Banach space valued functions which comprises spectral distributions

Figure 1: Theorem 6.10 and the arbitrary-coefficient question on page 12 of the source paper.

that product need not be a rational integer. The replacement is to put all coefficients in one number field K and take the field norm: the product is a nonzero algebraic integer, so its norm down to $\mathbb{Q}$ is a nonzero rational integer.

The other embeddings of K replace H_n by Galois-conjugate matrices. They need not be selfadjoint, but their entries and row sparsity are uniformly bounded, so all their eigenvalues stay in one fixed disk. The conjugate factors in the field norm therefore cost only $\exp(O(N_n^2))$. This gives exactly the scale needed to rule out a positive proportion of distinct eigenvalues collapsing into an arbitrarily small punctured interval.

A second elementary number-field observation handles large multiplicities. If an eigenvalue has normalized multiplicity at least $1/k$, its degree over K is at most k . Its absolute conjugates are uniformly bounded, so only finitely many such algebraic integers exist. Avoiding this finite set and using the discriminant estimate gives a punctured-neighborhood bound tending to zero. Portmanteau's inequalities then force convergence at every atom.

4 Main result

Theorem 1 (Algebraic-coefficient pointwise approximation). *In the set-up above, assume that every value in the finite local coefficient set S is a real algebraic number. Then, for every $x \in \mathbb{R}$,*

$$\lim_{n \rightarrow \infty} \nu_n(\{x\}) = \nu(\{x\}).$$

Consequently, if

$$F_n(t) = \nu_n((-\infty, t]) \quad \text{and} \quad F(t) = \nu((-\infty, t]),$$

then

$$\lim_{n \rightarrow \infty} \|F_n - F\|_\infty = 0.$$

Remark 1. The theorem strictly extends the integer hypothesis in source Theorem 6.10: it includes arbitrary real algebraic local values in a common number field. It does not include a coefficient set containing a transcendental number.

5 Proof

Multiplying every H_n by the same positive integer only rescales the spectral variable and does not affect either conclusion. Since S is finite, we may therefore choose a finite Galois extension $K/\mathbb{Q}$, realized inside $\mathbb{C}$ with its distinguished identity embedding, such that

$$S \subset \mathcal{O}_K \cap \mathbb{R}.$$

For $\sigma \in \text{Gal}(K/\mathbb{Q})$, let H_n^σ be obtained by applying σ entrywise. There is a constant $M \geq 1$, independent of n and σ , such that every row sum of absolute values of every H_n^σ is at most M . Indeed, the number of nonzero entries per row is bounded in terms of d, R , and there are only finitely many numbers $\sigma(s)$ with $s \in S$. Thus every eigenvalue of every H_n^σ has modulus at most M .

Lemma 1 (Only finitely many possible large atoms). *For every positive integer k , there is a finite set X_k of algebraic integers, depending only on K, M, k , such that*

$$\nu_n(\{\alpha\}) \geq \frac{1}{k} \implies \alpha \in X_k$$

for every n and every eigenvalue α of H_n .

Proof. Let α have multiplicity m as a root of $p_n(t) = \det(tI - H_n) \in \mathcal{O}_K[t]$. If q_α is its minimal polynomial over K , separability in characteristic zero gives $q_\alpha^m \mid p_n$. Therefore

$$m[K(\alpha) : K] \leq N_n.$$

If $m/N_n \geq 1/k$, then $[K(\alpha) : K] \leq k$. Moreover α is integral over $\mathcal{O}_K$, hence is an algebraic integer over $\mathbb{Z}$.

Every embedding of $\mathbb{Q}(\alpha)$ into $\mathbb{C}$ extends to an embedding of $K(\alpha)$. Its restriction to K is some σ , and the image of α is an eigenvalue of H_n^σ ; hence every absolute conjugate of α has modulus at most M . Finally,

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] \leq [K : \mathbb{Q}] k.$$

There are only finitely many algebraic integers of bounded degree whose conjugates all have bounded modulus: their monic minimal polynomials have bounded degree and bounded integer coefficients. Taking all such numbers as X_k proves the claim. $\square$

Lemma 2 (Number-field anti-clustering estimate). *Put $q = [K : \mathbb{Q}]$. Fix $x \in \mathbb{R}$ and $k \geq 1$. Choose $0 < \varepsilon < 1/2$ such that*

$$I_{x,\varepsilon} := (x - \varepsilon, x + \varepsilon) \setminus \{x\} \quad \text{satisfies} \quad I_{x,\varepsilon} \cap X_k = \emptyset.$$

Then, for every n ,

$$\nu_n(I_{x,\varepsilon}) \leq \left(\frac{q \log(2M)}{\log(1/(2\varepsilon))} + \frac{1}{k} \right)^{1/2}. \quad (1)$$

Proof. List the eigenvalues of H_n , with multiplicity, as $\lambda_1, \dots, \lambda_N$, where $N = N_n$. Define the nonzero-pair discriminant

$$D_n = \prod_{\substack{1 \leq i < j \leq N \\ \lambda_i \neq \lambda_j}} (\lambda_i - \lambda_j)^2.$$

It is a nonzero algebraic integer. It is also fixed by every automorphism of $\overline{\mathbb{Q}}$ fixing K : such an automorphism permutes the multiset of roots of p_n , preserves equality and inequality of roots, and the square removes the sign. Thus

$$D_n \in \mathcal{O}_K \setminus \{0\}.$$

Its field norm is a nonzero rational integer. For every nonidentity $\sigma \in \text{Gal}(K/\mathbb{Q})$, the conjugate $\sigma(D_n)$ is the analogous nonzero-pair product for H_n^σ . Since all conjugate eigenvalues have modulus at most M ,

$$1 \leq |N_{K/\mathbb{Q}}(D_n)| \leq |D_n|(2M)^{(q-1)N^2}.$$

Consequently,

$$|D_n| \geq (2M)^{-(q-1)N^2}. \quad (2)$$

Set

$$a = \nu_n(I_{x,\varepsilon}), \quad b = \nu_n(\{x\}), \quad s = \sum_{\lambda \in I_{x,\varepsilon}} \nu_n(\{\lambda\})^2.$$

No eigenvalue in $I_{x,\varepsilon}$ has mass at least $1/k$, by Lemma 1; hence

$$s < \frac{1}{k}. \quad (3)$$

The normalized number of *ordered* pairs of unequal eigenvalue occurrences which both lie in $(x - \varepsilon, x + \varepsilon)$ is exactly

$$(a + b)^2 - (b^2 + s) = a^2 + 2ab - s.$$

The square in D_n makes its factors correspond precisely to ordered unequal pairs. Factors belonging to the displayed collection have modulus at most 2ε , while all remaining factors have modulus at most $2M$. Therefore

$$|D_n| \leq (2\varepsilon)^{N^2(a^2+2ab-s)} (2M)^{N^2}. \quad (4)$$

Combining (2) and (4) gives

$$a^2 + 2ab - s \leq \frac{q \log(2M)}{\log(1/(2\varepsilon))}.$$

Since $b \geq 0$, estimate (3) now yields (1). □

Proof of Theorem 1. Fix $x \in \mathbb{R}$. For each k , Lemma 1 allows us to choose

$$0 < \varepsilon_k < \min\{1/k, 1/2\}, \quad I_{x, \varepsilon_k} \cap X_k = \emptyset.$$

Let

$$\delta_k = \left(\frac{q \log(2M)}{\log(1/(2\varepsilon_k))} + \frac{1}{k} \right)^{1/2}.$$

Then $\delta_k \rightarrow 0$ and Lemma 2 says

$$\sup_n \nu_n(I_{x, \varepsilon_k}) \leq \delta_k. \quad (5)$$

Weak convergence and the Portmanteau theorem give

$$\limsup_{n \rightarrow \infty} \nu_n(\{x\}) \leq \nu(\{x\}). \quad (6)$$

On the other hand, for the open interval $J_k = (x - \varepsilon_k, x + \varepsilon_k)$,

$$\begin{aligned} \nu(\{x\}) &\leq \nu(J_k) \leq \liminf_{n \rightarrow \infty} \nu_n(J_k) \\ &\leq \liminf_{n \rightarrow \infty} \nu_n(\{x\}) + \delta_k, \end{aligned}$$

where the last inequality follows from (5) by taking a subsequence realizing the lower limit of the singleton masses. Letting $k \rightarrow \infty$ and combining with (6) proves

$$\nu_n(\{x\}) \rightarrow \nu(\{x\}).$$

Finally, the standard distribution-function lemma says that weak convergence of probability measures together with convergence of the mass of every singleton is equivalent to uniform convergence of their cumulative distribution functions. For completeness, one proves it by isolating the finitely many atoms of ν larger than a prescribed tolerance, choosing a finite partition by continuity points of ν on the remaining intervals, and using monotonicity between consecutive partition points. The atom convergence controls the isolated jumps. Applying the lemma to $\nu_n \Rightarrow \nu$ completes the proof. $\square$

6 Verification notes and limitations

Proof audit. The argument was checked independently at the following structural points: (i) one denominator clears the finite coefficient set; (ii) multiplicity bounds relative degree over K ; (iii) the squared nonzero-pair discriminant belongs to $\mathcal{O}_K$ despite repeated eigenvalues; (iv) the field norm and the row-sum bounds on every conjugate matrix give (2); and (v) the Portmanteau inequalities have the required directions. No numerical experiment is used as proof evidence.

Human-review focus. The most important line to verify is $D_n \in \mathcal{O}_K$ in Lemma 2. Once this is accepted, all constants are uniform in n , and the remaining argument follows the standard discriminant anti-clustering scheme.

Scope. The original arbitrary-real-coefficient problem remains open. The proof uses algebraicity exactly through the nonzero field-norm lower bound and the finiteness of bounded-house algebraic integers of bounded degree. Neither input is available for a general transcendental coefficient tuple.

7 Novelty check

The exact graph-sequence formulation was not located in a bounded search. The search covered the local run indexes; the phrases *algebraic coefficients Lück approximation spectral measures graph sequences*, *number field Lück approximation*, and *arbitrary coefficients graph sequences spectral distribution*; and arXiv:1304.0945, arXiv:math/0701294, the Abért–Thom–Virág preprint cited in the source, and arXiv:2603.01610.

The number-field mechanism is not itself new: Thom proves the algebraic coefficient analogue for sofic group approximations [4]. The 2026 work of Donoso-Echenique, Pogorzelski, and Schrödl-Baumann proves pointwise convergence for finitely many rational values in a random-operator/sofic-group setting and treats arbitrary complex coefficients through adapted approximants with varying rational coefficients [2]. No searched source stated Theorem 1 for every weakly convergent bounded-degree graph sequence. Accordingly, novelty of this extension is *provisional*, not established.

References

- [1] M. Abért, A. Thom, and B. Virág, *Benjamini–Schramm convergence and pointwise convergence of the spectral measure*, circulated preprint, 2013.
- [2] M. Donoso-Echenique, F. Pogorzelski, and M. Schrödl-Baumann, *Random operators, spectral measures, and local empirical convergence in sofic groups*, arXiv:2603.01610, 2026.
- [3] F. Pogorzelski, *Convergence theorems for graph sequences*, International Journal of Algebra and Computation 24 (2014), 1233–1251; arXiv:1304.0945.
- [4] A. Thom, *Sofic groups and diophantine approximation*, Communications on Pure and Applied Mathematics 61 (2008), 1155–1171; arXiv:math/0701294.